

★ Example of sliding window.

set : $x = \begin{bmatrix} [1, 2, 3, 4] \\ [5, 6, 7, 8] \\ [9, 10, 11, 12] \\ [13, 14, 15, 16] \end{bmatrix}$

kernel size $KH = KW = 2$ it selects a small 2×2 window each time.

How should I select the first ~~small~~ ^{$(0,0)$} small window?

The ~~top-left~~ top-left corner of the window is at $(0,0)$

So select $x[0:0+2, 0:0+2] \rightarrow x[0:2, 0:2]$

result is

$\rightarrow \begin{bmatrix} [1, 2] \\ [5, 6] \end{bmatrix}$ $\rightarrow x[\text{row_start} : \text{row_end}, \text{col_start} : \text{col_end}]$

move the window one step to the right $\rightarrow$

The top-left corner of second window becomes $(0,1)$.

$x[0:2+0, 1:2+1] \rightarrow x[0:2, 1:3] \rightarrow \begin{bmatrix} 2, 3 \\ 6, 7 \end{bmatrix}$

Continue to go right. $X[0:2, 2:4] \rightarrow \begin{bmatrix} 3, 4 \\ 7, 8 \end{bmatrix}$

So there are 3 windows in the first line.
Now the window slides down one line. Now the top-left corner is at (1, 0).

$X[1:3, 0:2] \rightarrow \begin{bmatrix} 5, 6 \\ 9, 10 \end{bmatrix}$ continue $X[1:3, 1:3]$

$\rightarrow \begin{bmatrix} 1, 7 \\ 10, 11 \end{bmatrix}$ continue $X[1:3, 2:4]$ $\begin{bmatrix} 7, 8 \\ 11, 12 \end{bmatrix}$

So how many windows in total?

Input size 4×4
kernel 2×2

$$H_{out} = \frac{H + 2P - KH}{S} + 1 = \frac{4 + 0 - 2}{1} + 1 = 3$$

padding $p=0$
stride $=1$

$$W_{out} = \frac{W + 2P - KW}{S} + 1 = \frac{4 + 0 - 2}{1} + 1 = 3$$

So there are 9 small windows in total, these correspond to the 9 locations in the output feature map.

AA don't include the right boundary

e.p: $[2, 4] \rightarrow [2, 3]$ ✓

EinSum

$$\text{result} = \sum_h \sum_w \text{window}[h, w] * \text{kernel}[h, w]$$

e.p: window = np.array([

kernel = np.array([

[1, 2],

[1, 0],

[5, 6]

[0, 1]

])

])

$$\text{result} = \text{NP. Sum}(\text{window} * \text{kernel})$$

$$= \begin{bmatrix} \begin{bmatrix} 1 \times 1 \\ 5 \times 0 \end{bmatrix}, \begin{bmatrix} 2 \times 0 \\ 6 \times 1 \end{bmatrix} \end{bmatrix} = \begin{bmatrix} [1, 0] \\ [0, 6] \end{bmatrix}$$

$$1 + 0 + 0 + 6 = 7 \quad \text{so finally output result} = 7$$